



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, March 13, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

FEM intro

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Introduction to Mechanics

How does a car resist vibration when driving on rough roads?



How can the design of a drone be optimized?



What's the performance of a cellphone in a drop test?



How can spine stress be released by an operation?



To answer these questions and find solutions, we must rely on **Mechanics**!



Introduction to Mechanics

Mechanics is a study of :

How things work,
how the world works!

- Automobiles
- Buildings
- Electronics
- Biology
- Packaging
- Consumer Goods



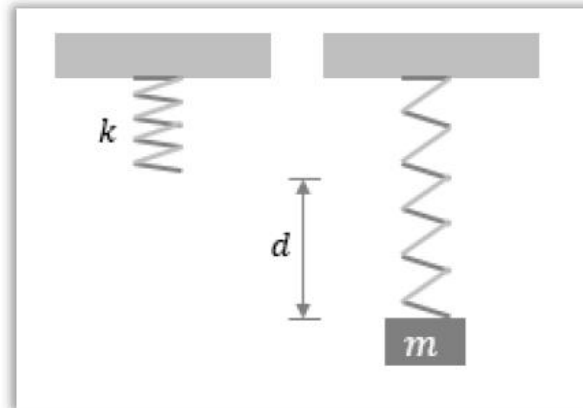
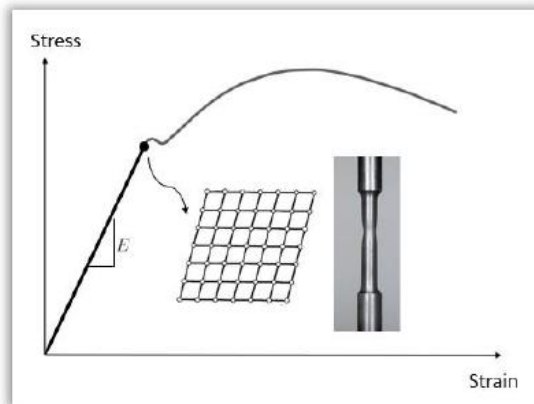
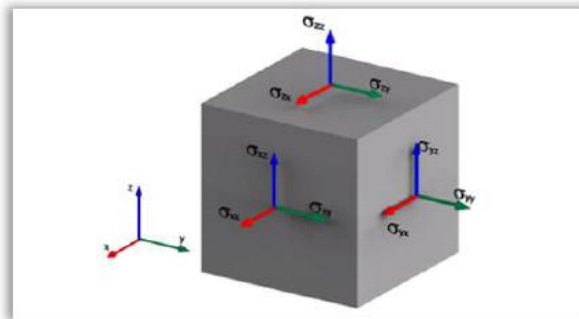
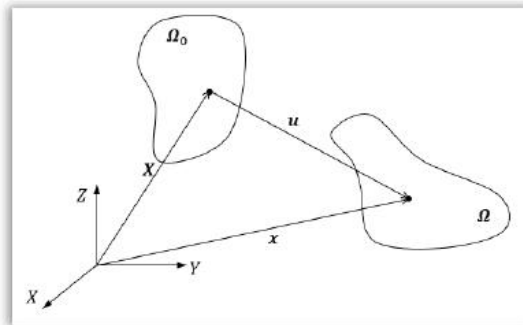
Introduction to Mechanics

This course is on engineering simulation. It provides the start of the necessary background of physics and numerical methods for every engineer to understand and use simulation.

Physics



Simulation

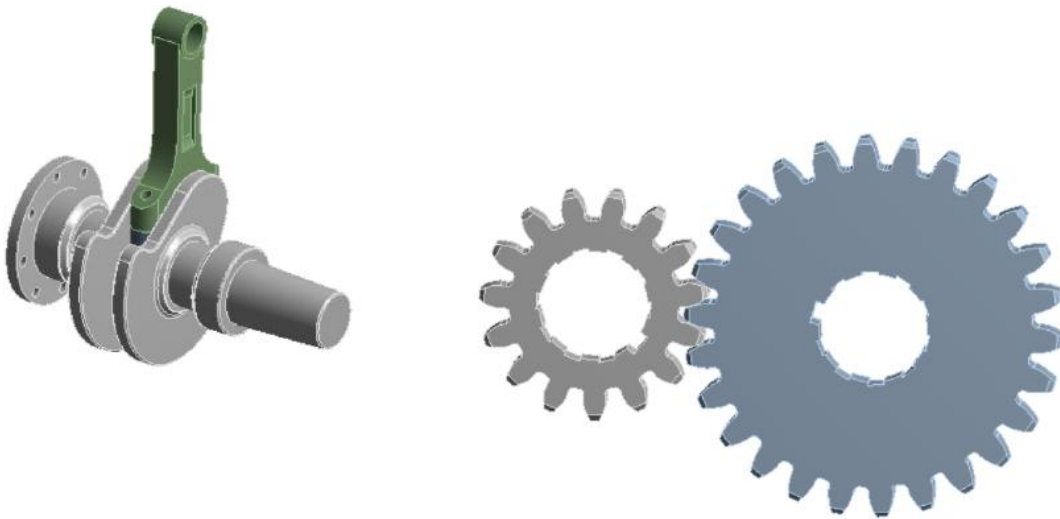


Rigid Body and Flexible Body

In mechanics, we can treat a body as a rigid body or a flexible one. Continuum mechanics deals with material deformation, so it focuses on flexible bodies.

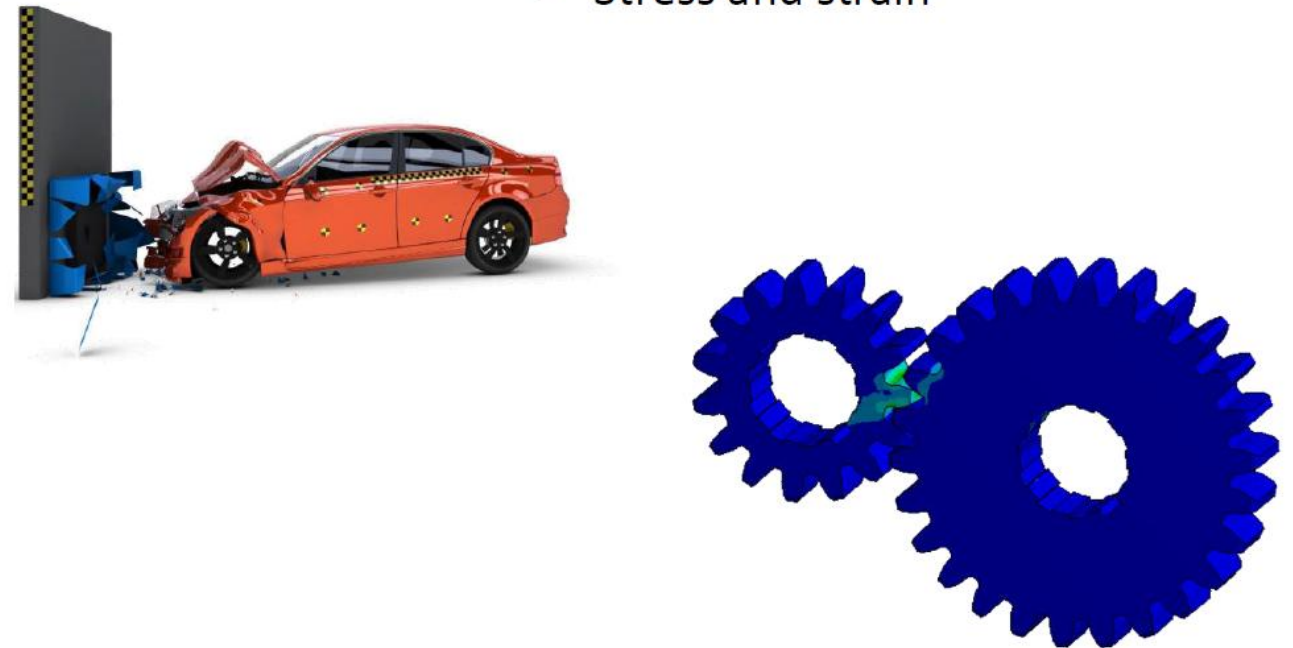
Rigid body

- Rigid motion
- Reaction force



Flexible body

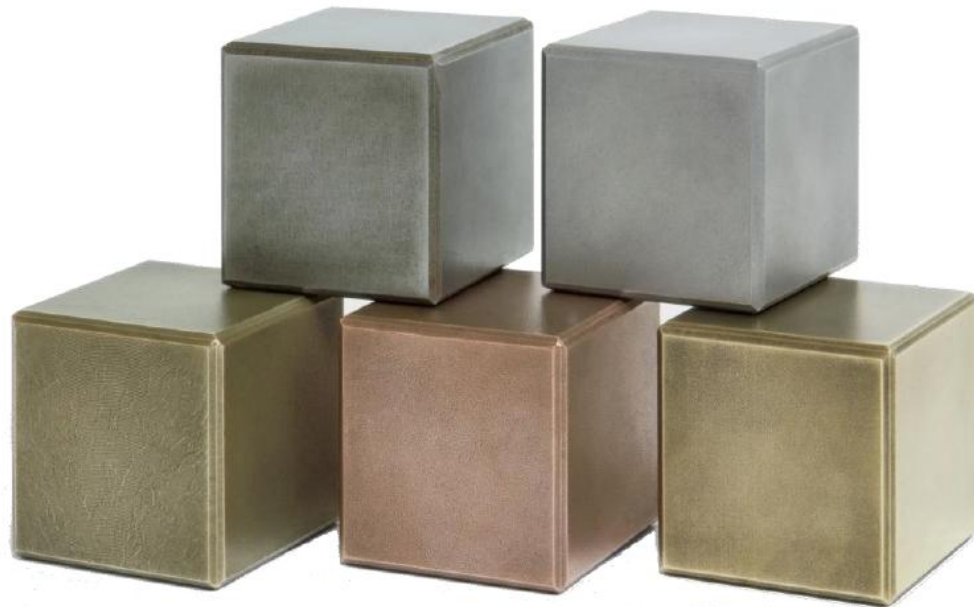
- Deformation
- Stress and strain



Classical Mechanics

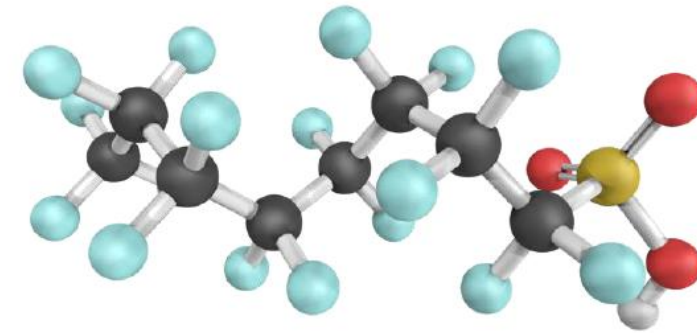
In this course, we are going to learn about the macroscopic behavior of a structure or material. We are looking for the motion, deformation, stress and strain of a structure as a holistic body.

Continuum Mechanics



Materials are treated as continuous mass instead of discrete particles/atoms.

Molecular Mechanics



The study of microstructure is called molecular or quantum mechanics, which falls outside the scope of this course.



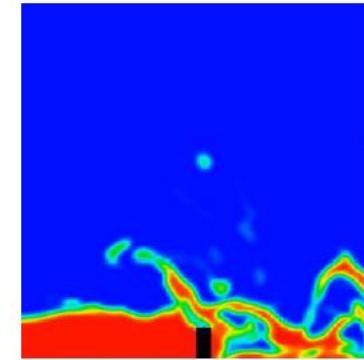
Continuum Mechanics

Two types of continuum states exist in mechanics: fluids (liquids and gases) and solids. Here we will focus on solid mechanics.

Continuum Mechanics

Fluid
Mechanics

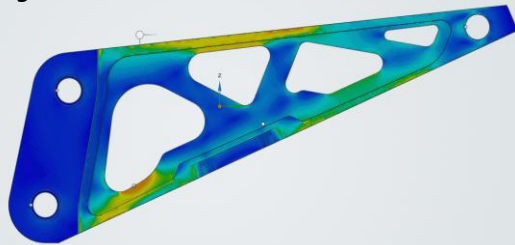
Solid
Mechanics



Physical principles

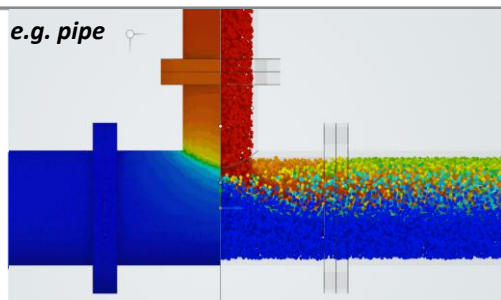
Solid Mechanics

e.g. bracket



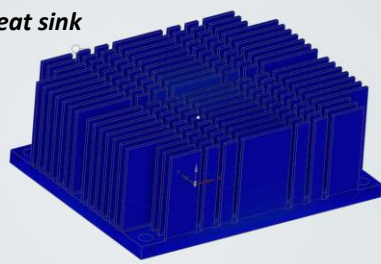
Fluid Mechanics

e.g. pipe



Thermal Analysis

e.g. heat sink



⋮

Physical principles

Governing equations

SM

Euler-Bernoulli beam deflection

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 w}{dx^2} \right) = q$$

FM

Navier-Stokes motion of viscous fluids

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \nu \nabla^2 \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{g}$$

TA

Fourier's law of heat transfer

$$q_x = -k \frac{dT}{dx}$$

Boundary conditions

SM

Fixed end

$$w|_{x=0} = 0; \frac{\partial w}{\partial x}|_{x=0} = 0$$

FM

Parallel flow

$$\frac{d^2 u}{dy^2} = -1; u(0) = u(1) = 0$$

TA

Insulation at $x=0$

$$-k \frac{dT}{dx} \Big|_{x=0} = 0$$

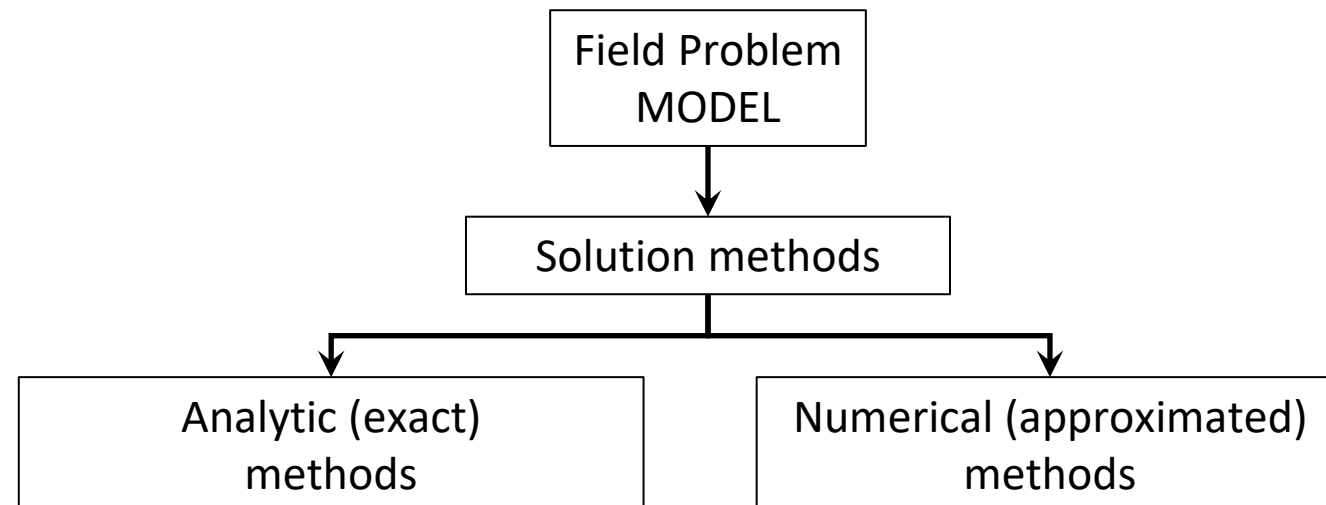
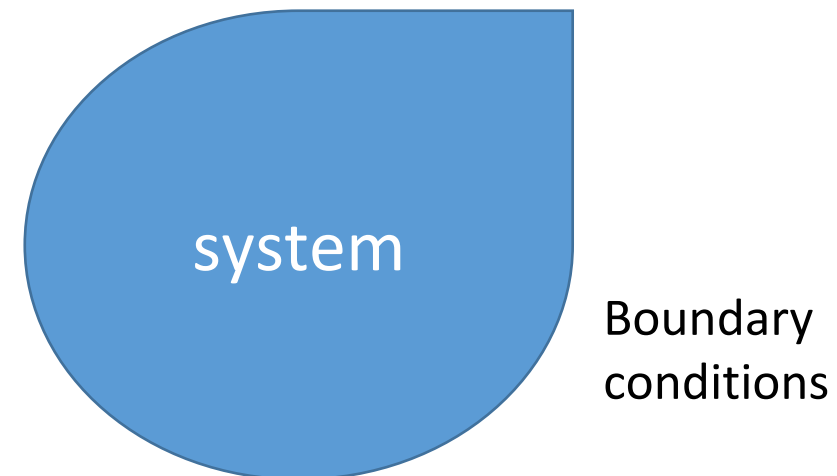


Some definitions...

FIELD PROBLEM: Given a certain system, with defined volume and boundary conditions, find the spatial distribution of one or more dependent variables.

Examples:

- Displacement, stress, strain
- Temperature (heat transfer)
- Electric charge
- Magnetic field



FEM

Historical Note

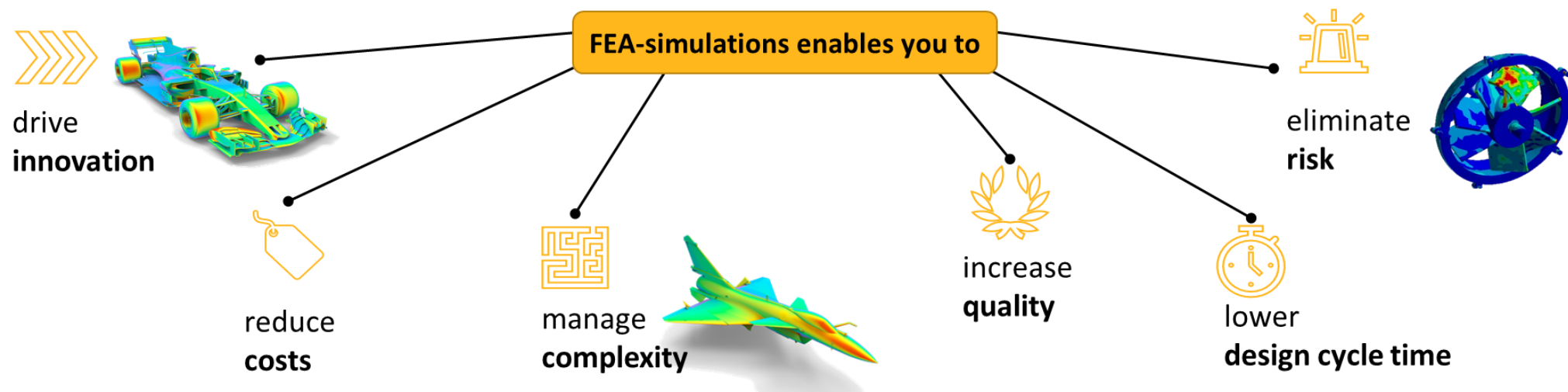
- *The finite element method of structural analysis was created by academic and industrial researchers during the 1950s and 1960s.*
- *The underlying theory is over 100 years old, and was the basis for pen-and-paper calculations in the evaluation of suspension bridges and steam boilers.*



Numerical Analysis – Finite Element Analysis (FEA) and Method (FEM)

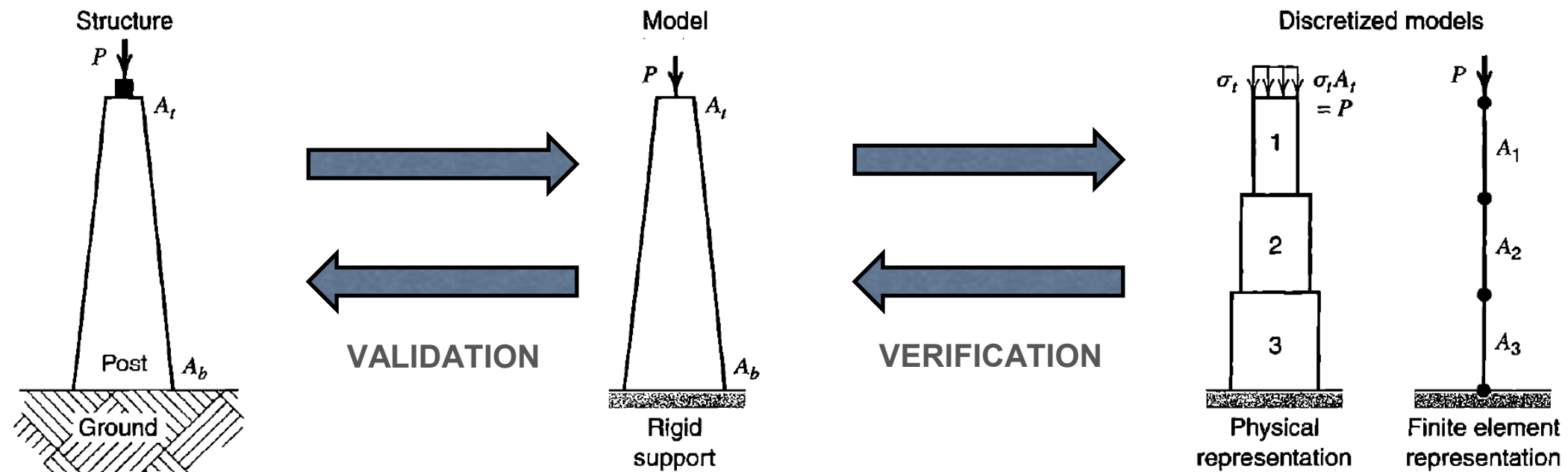
- FEM is the **mathematical method** used to conduct FEAs, in which partial differential equations are solved to **describe physical phenomena**.
- FEA are performed using FEM, which are implemented in **CAE** (computer-aided engineering) tools, providing a solution for **virtual prototyping**.
- FEA helps to solve **structural, heat transfer, fluid flow or electromagnetic problems** quickly by exploiting computational power and make **design changes** early product development stage.

Finite Element Analysis	Finite Element Method
Type of analysis	Numerical technique



FEA is powerful and widely used in many industries. It is fast (faster than prototyping) and can enable innovation that would be tedious or impossible with any other method, **IF USED CORRECTLY**.

Approximated nature of FEM



VERIFICATION: confirm that FEM calculations correctly represent mathematical model/equations on which it is based. **SOFTWARE COMPANIES CHECK THIS.**

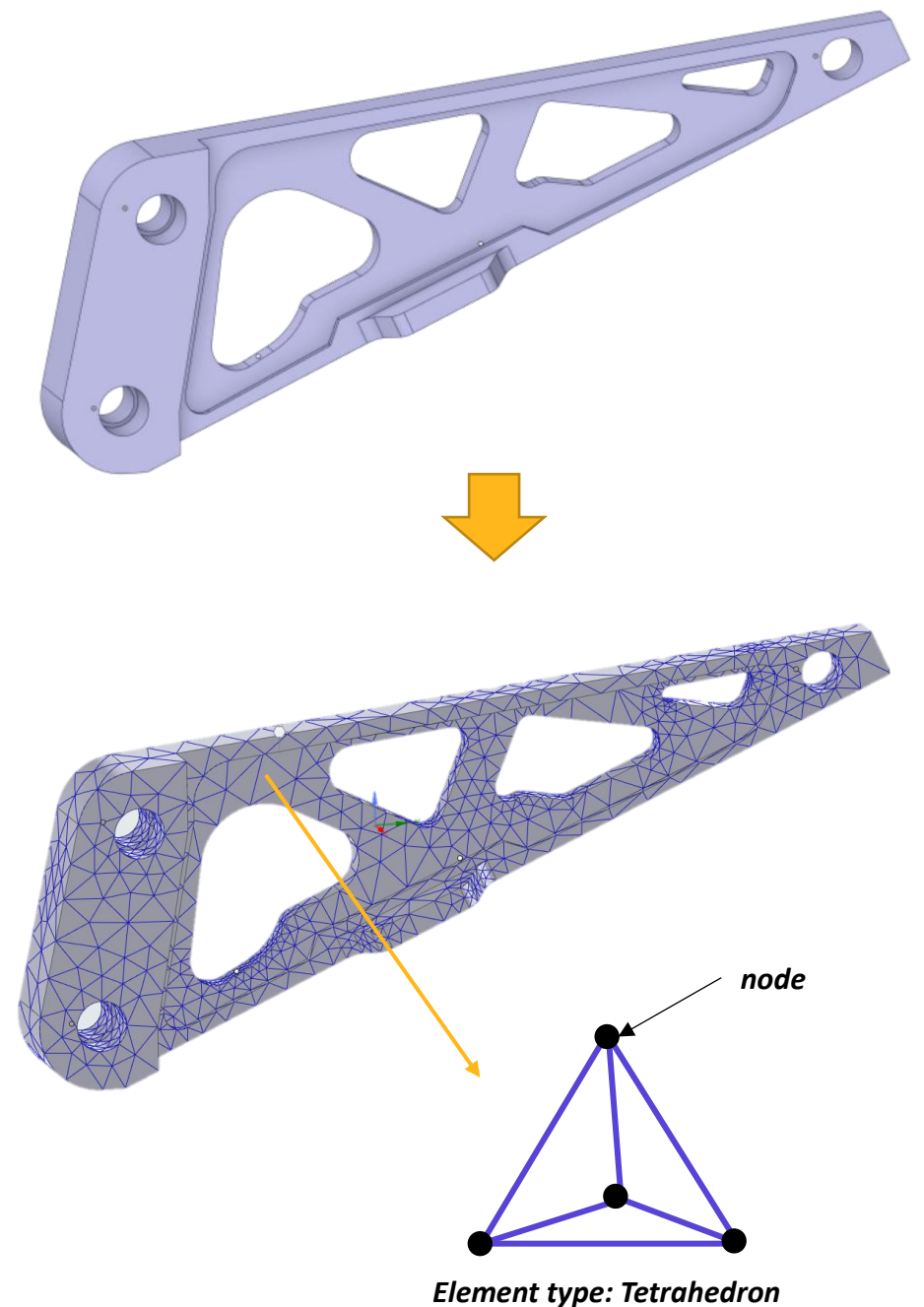
VALIDATION: confirm that the mathematical model you have solved with FEM is an accurate representation of the real physical system. **USER HAS TO DO THIS.** Validation is usually done by comparison of FEM results to experiments or simple hand calculations.

The essence of FEA is approximation by piecewise interpolation of a field quantity.

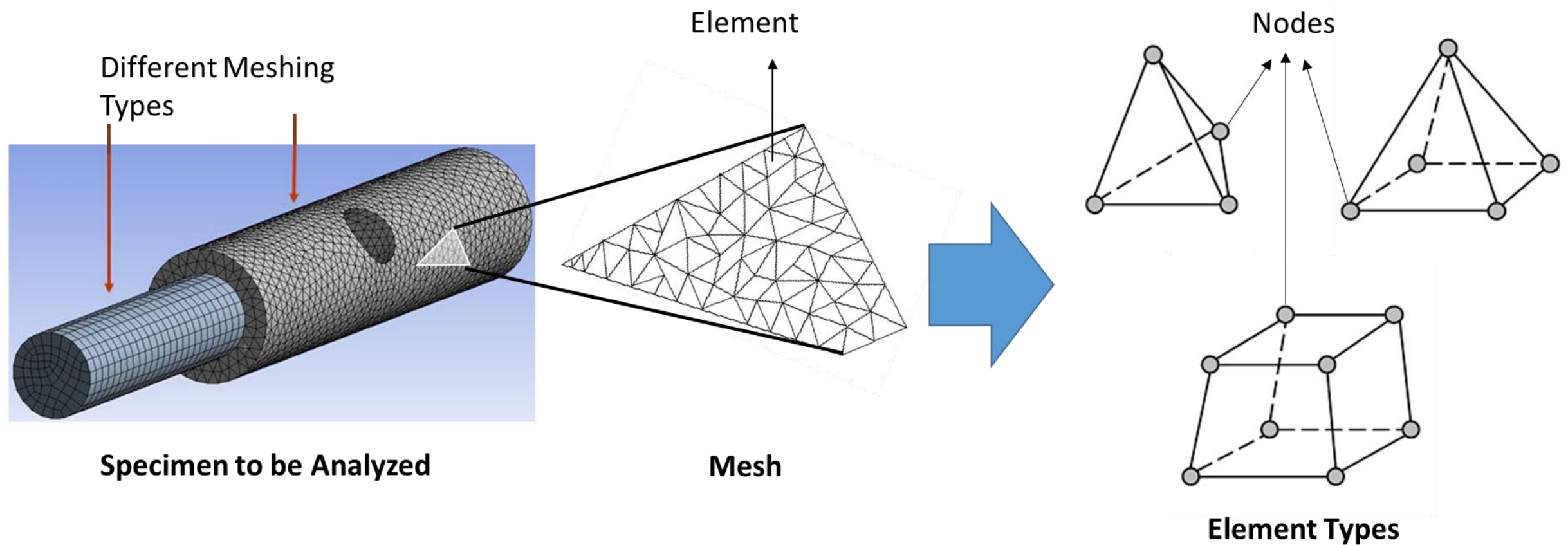


...other definitions

- FEM is a **numerical method** that exploits computational power to calculate solutions for a range of different engineering problems, offering **good approximate solution** to complex problems in the field of e.g. **solid mechanics** (e.g., statics and dynamics).
- At the core of FEM lies the **discretization**, which means a model is divided into a **finite** number of smaller **bodies or elements** that are connected at common points, the so-called **nodes**.
- Each node has a limited number of **degrees of freedom** (DoFs), so e.g. a continuum solid becomes discretized into a finite number of DoFs, based on the number and type of elements.
- For each of those elements a set of **algebraic equations** are solved in an **iterative fashion**.



...other definitions



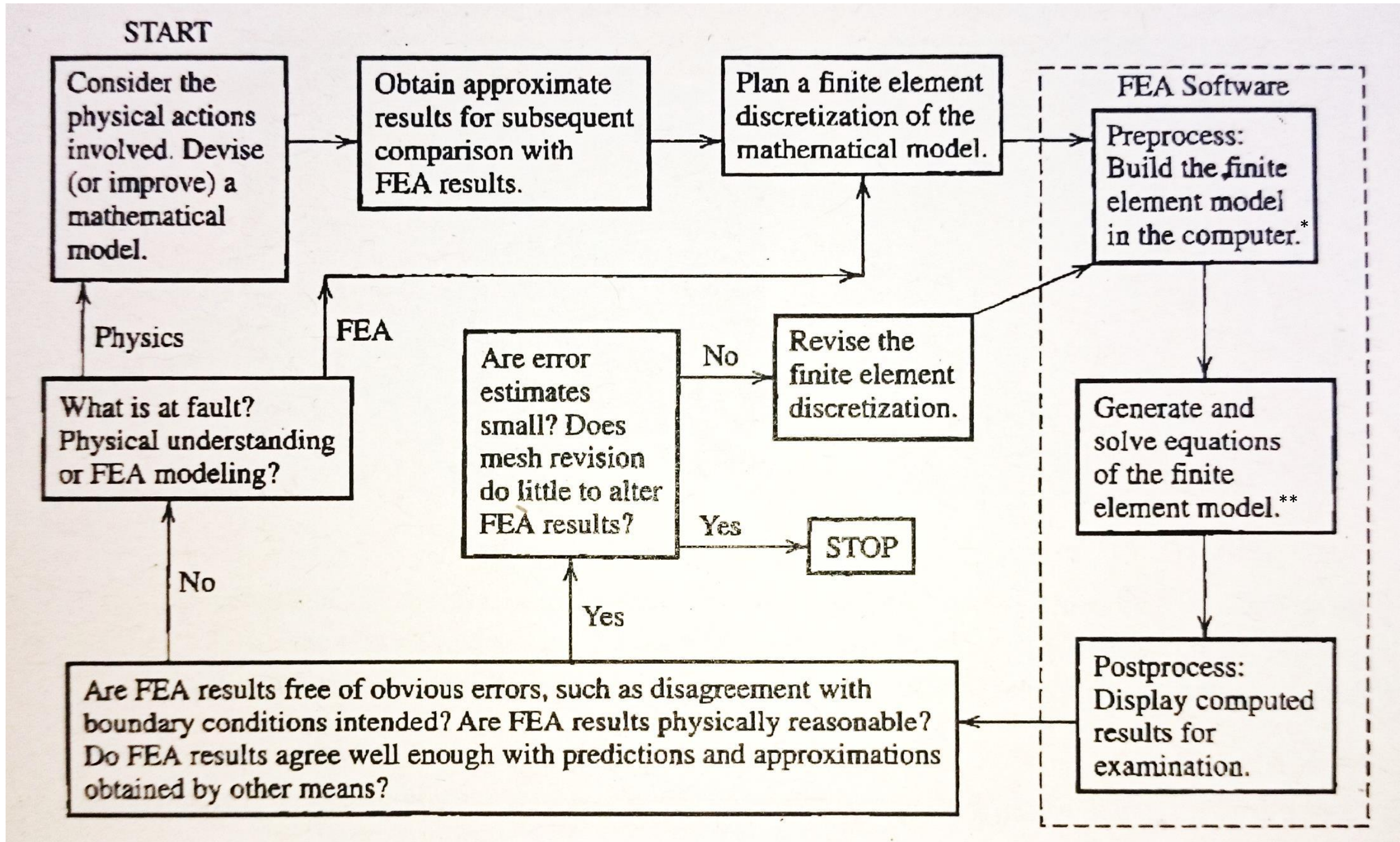
Meshing: Grid generation for dividing into elements.

Elements: Unit of the mesh.

Node: Solution points of the element



FEA workflow

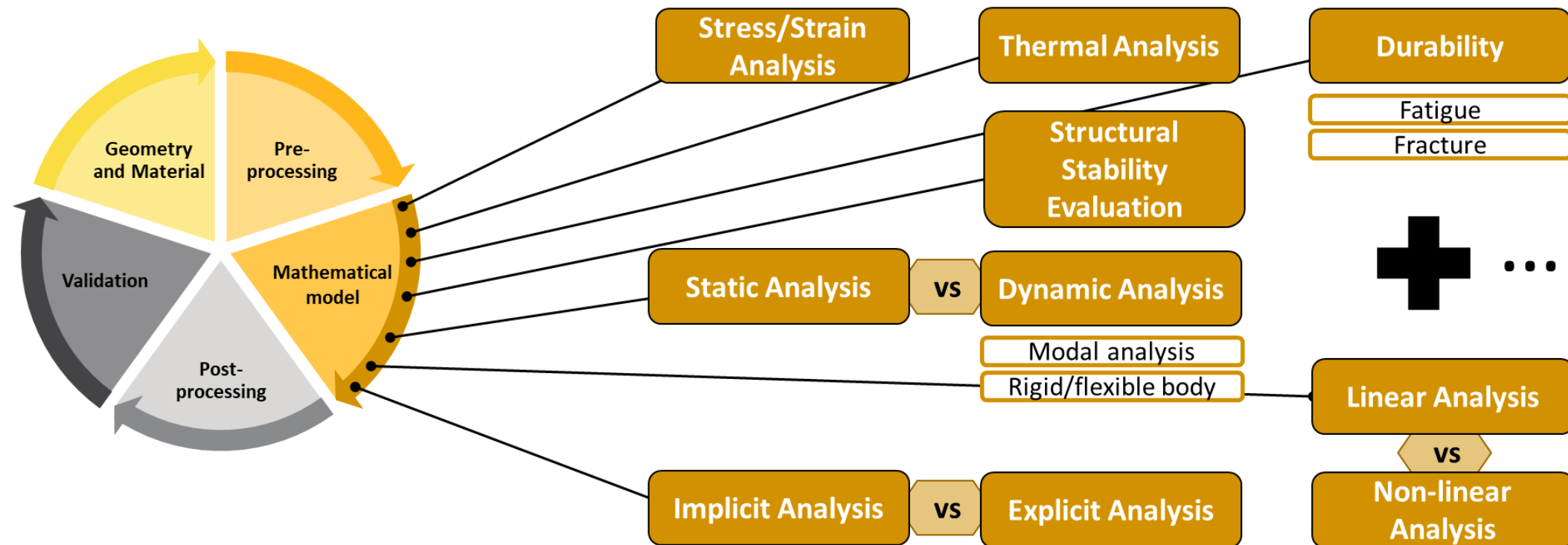
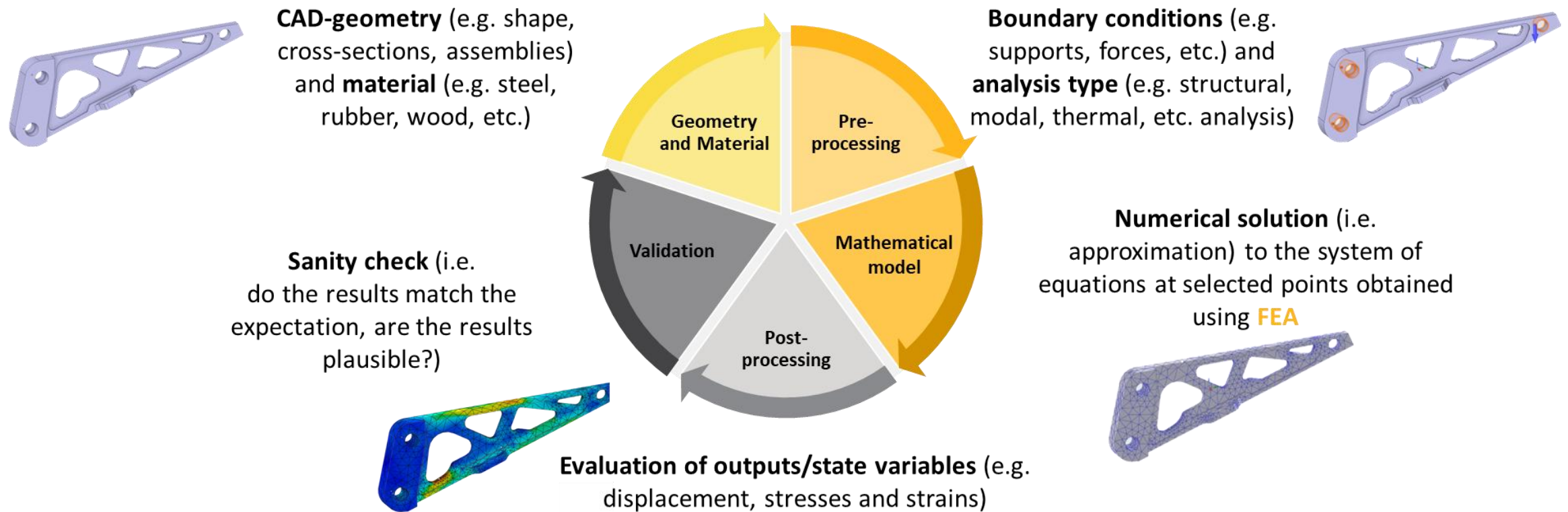


* Implementing simplifications (e.g. symmetries) if possible

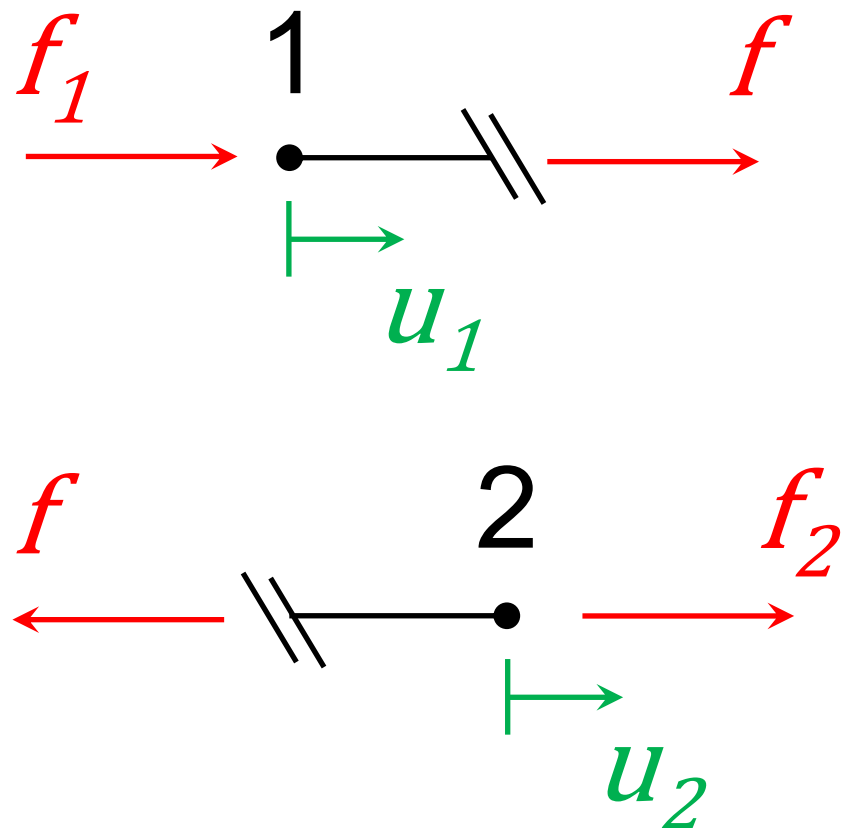
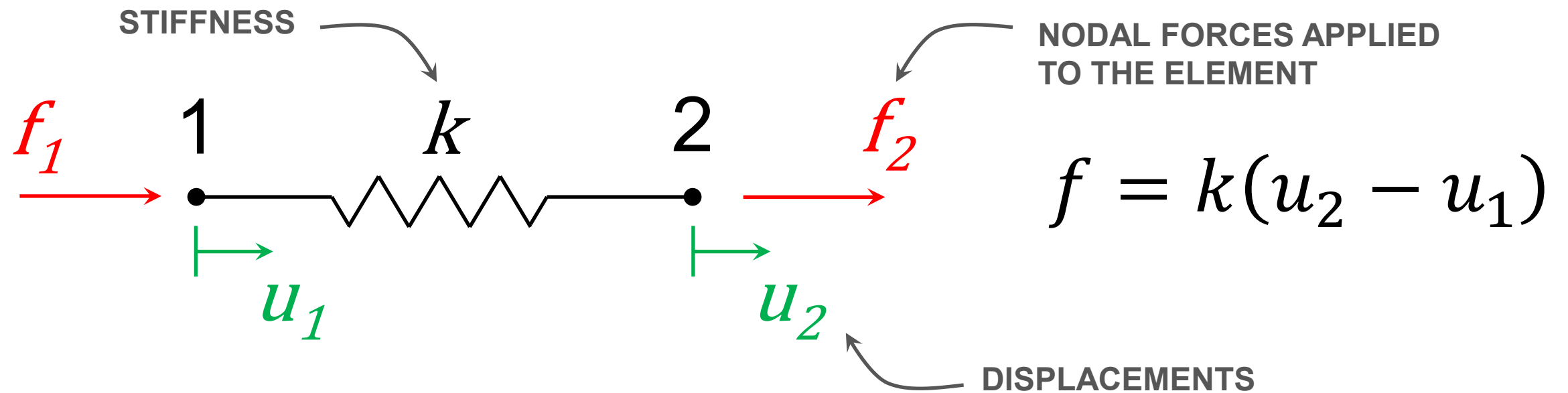
** Defining boundary conditions



FEA workflow



1D bar element example



$$\begin{cases} f_1 + k(u_2 - u_1) = 0 \\ f_2 - k(u_2 - u_1) = 0 \end{cases}$$

$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix}$$

$$[K] \{U\} = \{F\}$$



Loads and constraints

*Equation solved
for a static
analysis*

$$[K]\{U\} = \{F\}$$

External forces

Displacement vector

Stiffness matrix

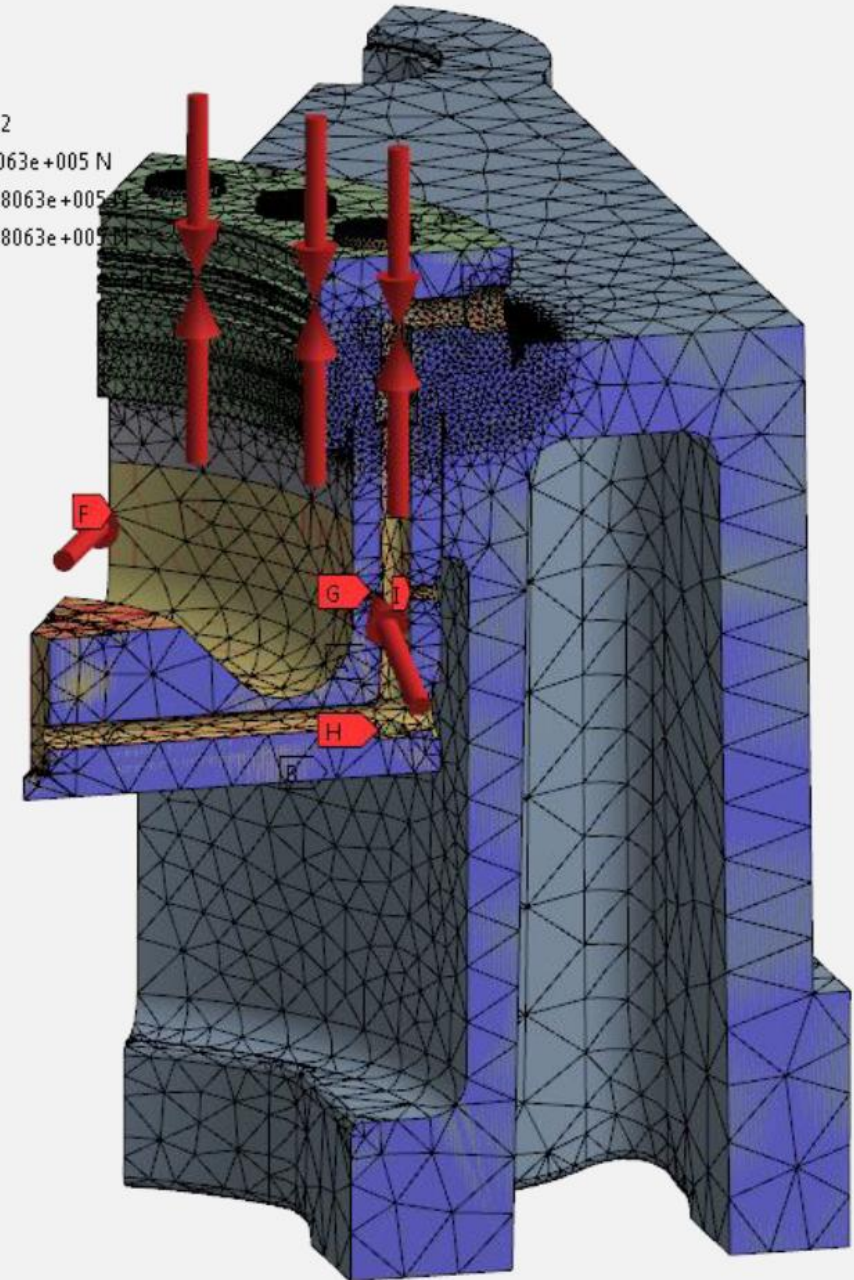
A: Modell Abheben des Zylinders von der Traverse Schrb, RandBedingungen Last 30MPa Schrb 10.9 minimale Vorspannung

RandBedingungen Last 30MPa Schrb 10.9 minimale Vorspannung

Time: 1. s

6/14/2011 1:33 PM

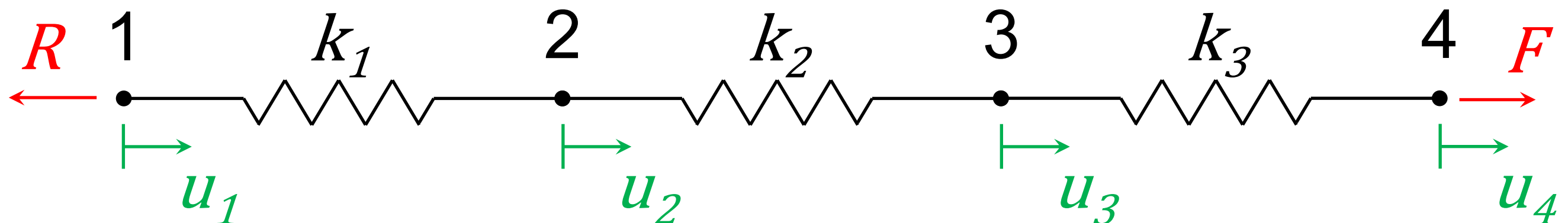
- A** Reibungsfreie Lagerung
- B** Reibungsfreie Lagerung 2
- C** Statische Spannung: 4.8063e+005 N
- D** Statische Spannung 2: 4.8063e+005 N
- E** Statische Spannung 3: 4.8063e+005 N
- F** Druck: 0. MPa
- G** Druck 2: 0. MPa
- H** Kraft: 0. N
- I** Kraft 2: 0. N
- J** Kraft 3: 0. N



The stiffness matrix

Assembly

$$\begin{cases} -R + k_1(u_2 - u_1) = 0 \\ -k_1(u_2 - u_1) + k_2(u_3 - u_2) = 0 \\ -k_2(u_3 - u_2) + k_3(u_4 - u_3) = 0 \\ -k_3(u_4 - u_3) + F = 0 \end{cases}$$



$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} -R \\ 0 \\ 0 \\ F \end{Bmatrix}$$

GLOBAL STIFFNESS MATRIX

$[K]$

$$\{U\} = \{F\}$$

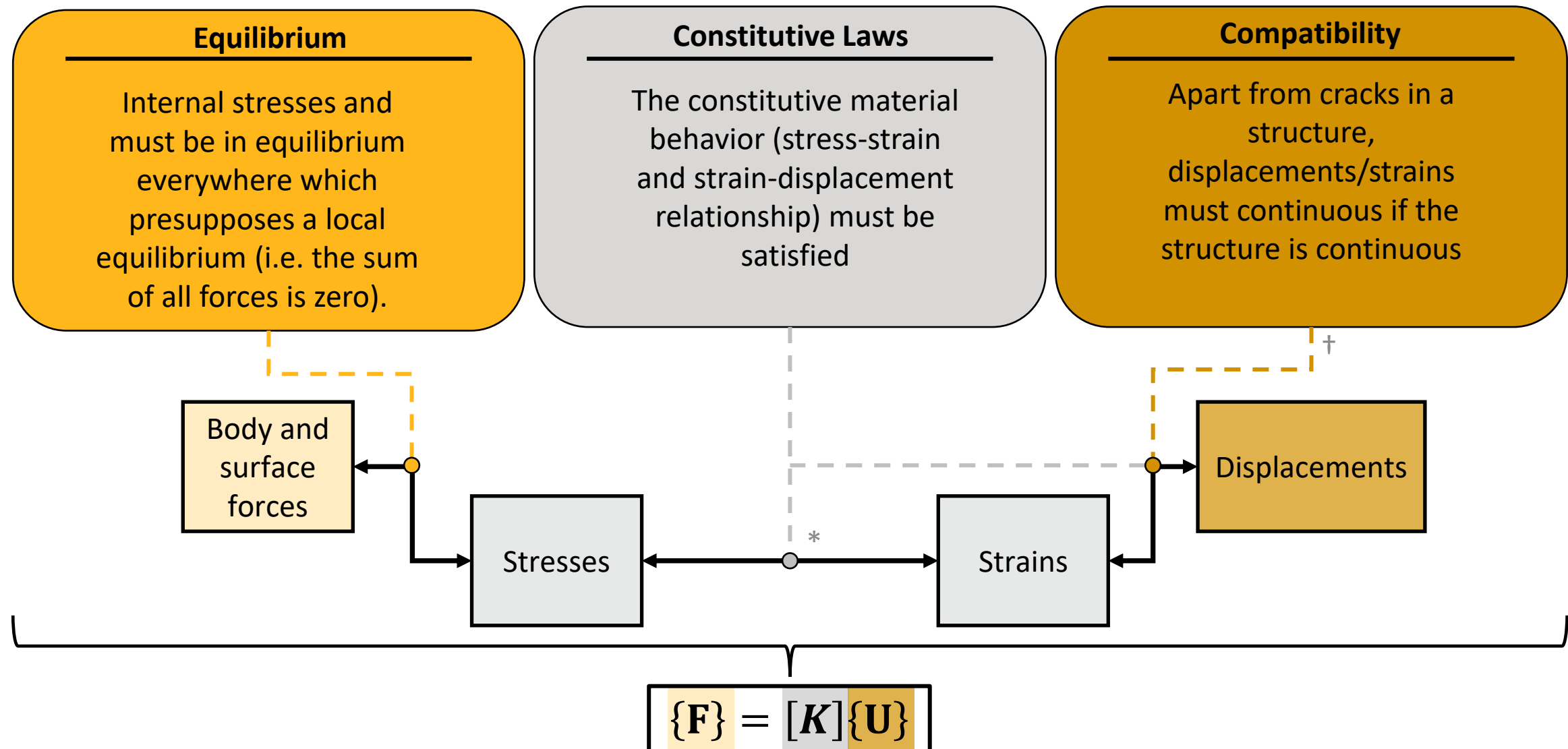


FEA process

1. Set model and elements
2. Develop element stiffness matrix, $[k]$
3. Assembly to form global stiffness matrix, $[K]$
4. Apply loads to nodes
5. Apply boundary conditions to supports
6. Solve for $\{U\}$ and $\{F\}$
7. Compute gradients ($\varepsilon = u,x$) and related quantities ($\sigma = E\varepsilon$)



FEA process



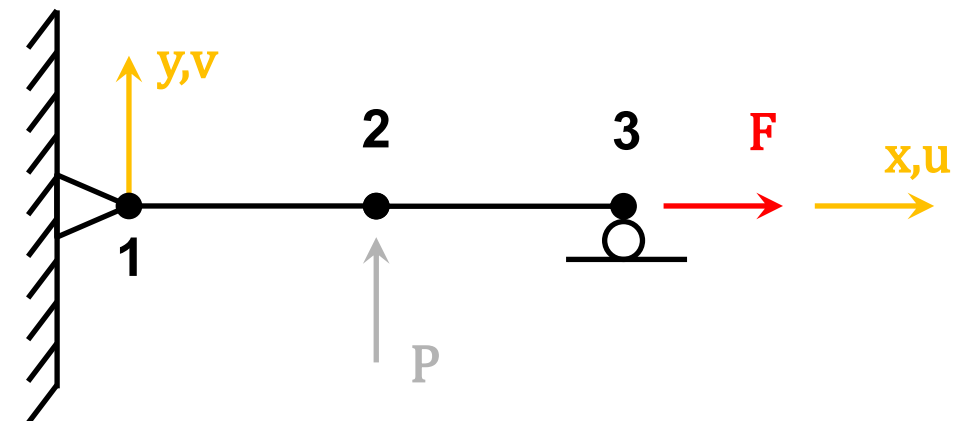
Note: Capital letters generally refer to the global and lowercase letters to the element matrix.



The stiffness matrix

Properties of stiffness matrix

1. Nonnegative k_{ij}
2. Symmetry
3. Sparsity
4. Singularity
(no or inadequate support)



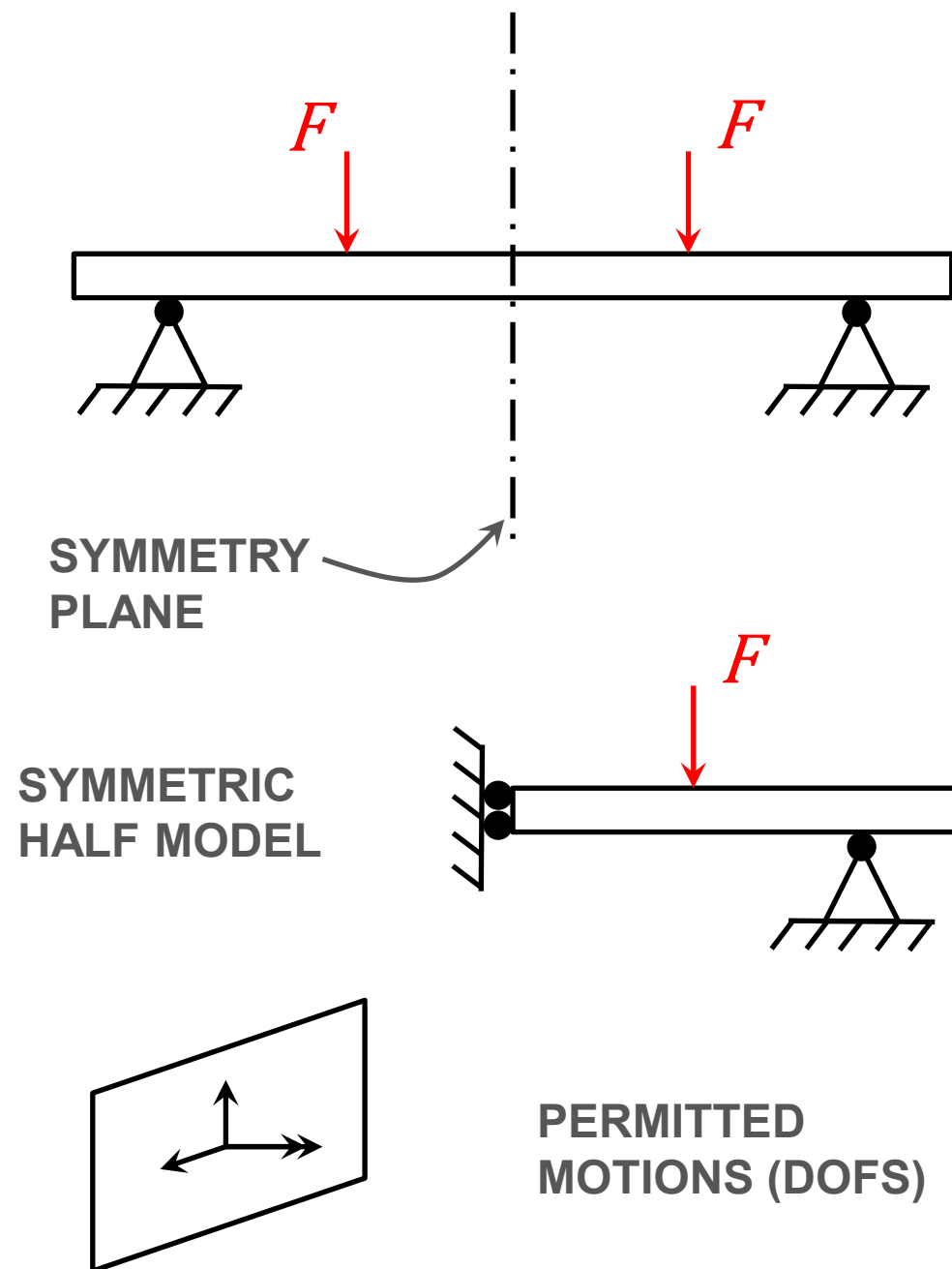
#1			#2			#3			
1	2	3	4	6	7	8	9	10	
0.260	0.000	-0.390	-0.260	0.000	-0.390	0.000	0.000	0.000	1
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	2
-0.390	0.000	0.785	0.390	0.000	0.385	0.000	0.000	0.000	3
-0.260	0.000	0.390	6.927	0.000	0.390	-6.667	0.000	0.000	4
0.000	0.000	0.000	0.000	0.096	0.240	0.000	-0.096	0.240	6
-0.390	0.000	0.385	0.390	0.240	1.585	0.000	-0.240	0.400	7
0.000	0.000	0.000	-6.667	0.000	0.000	6.667	0.000	0.000	8
0.000	0.000	0.000	0.000	-0.096	-0.240	0.000	0.096	-0.240	9
0.000	0.000	0.000	0.000	0.240	0.400	0.000	-0.240	0.800	10
									#1
									#2
									#3



Symmetries

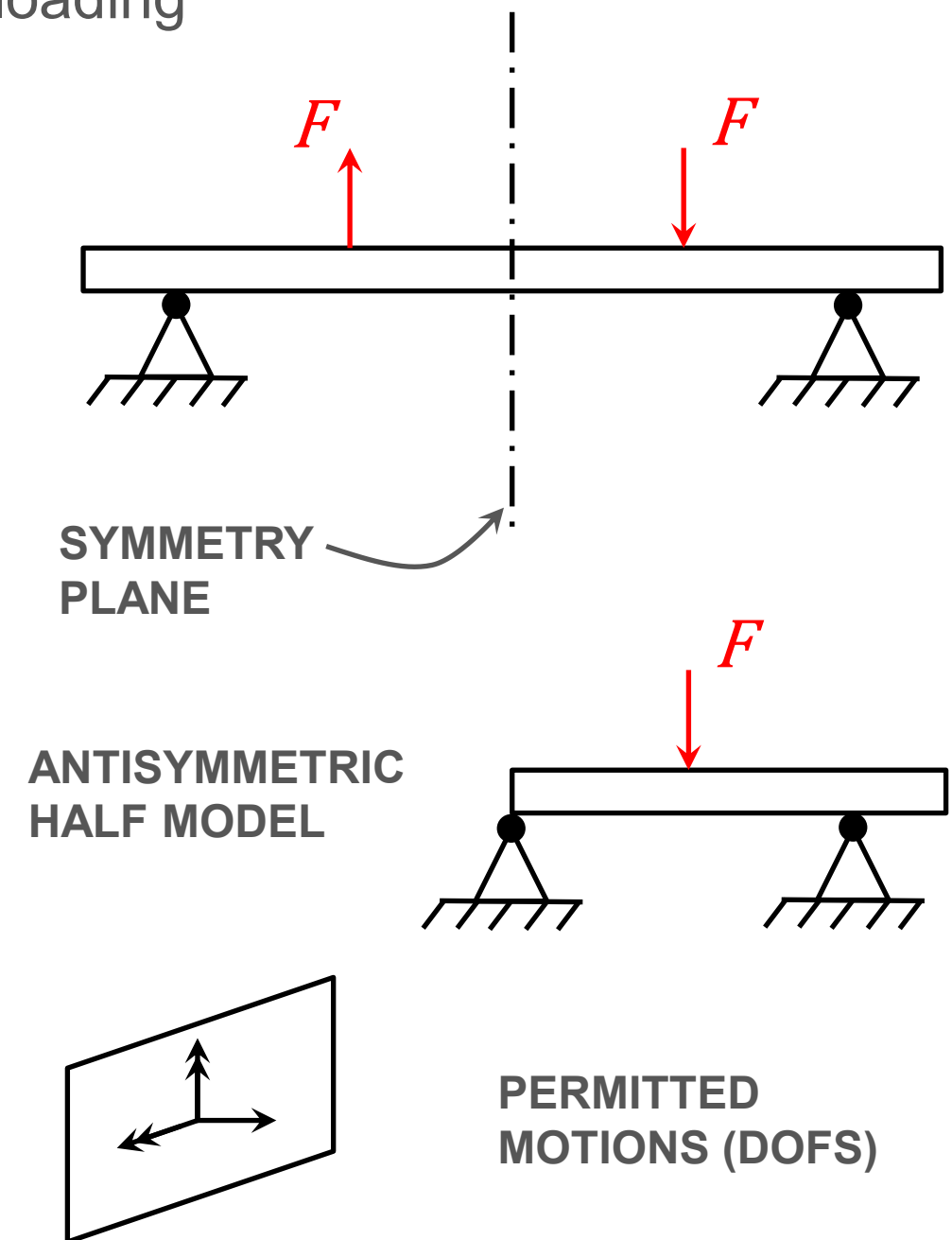
Reflective or mirror symmetry

Symmetric geometry and loading

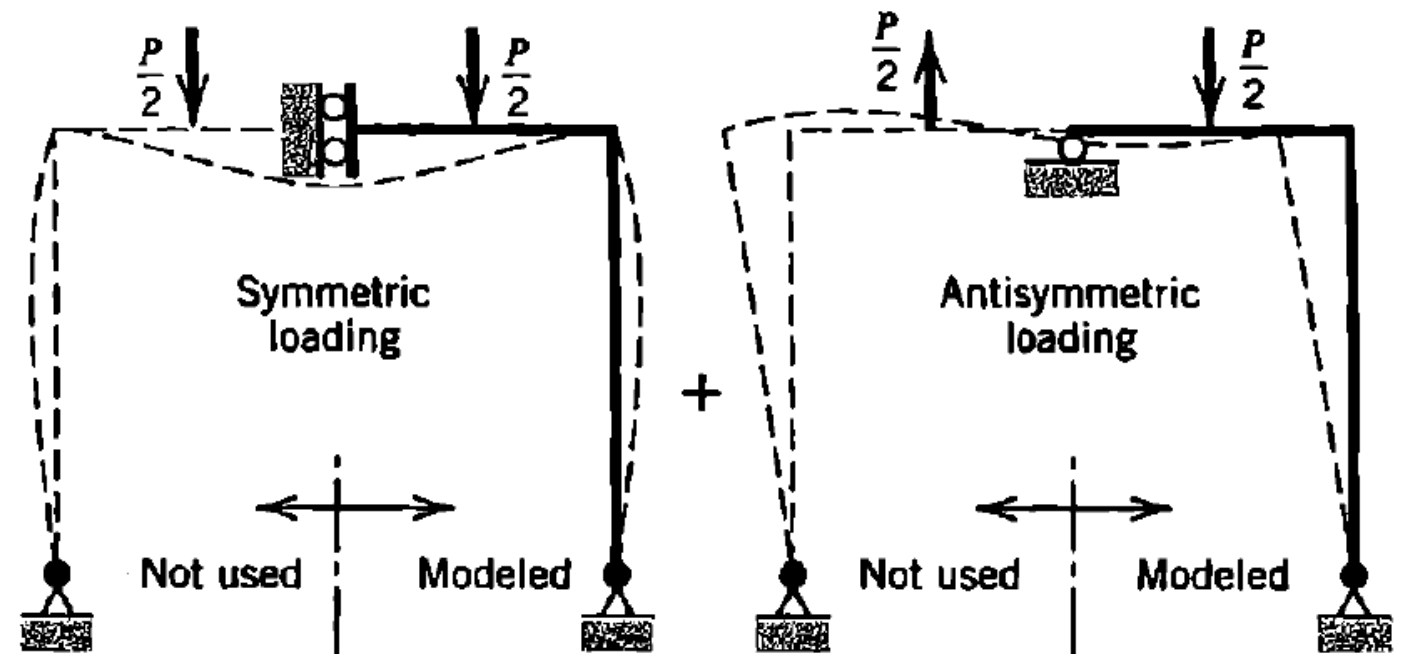
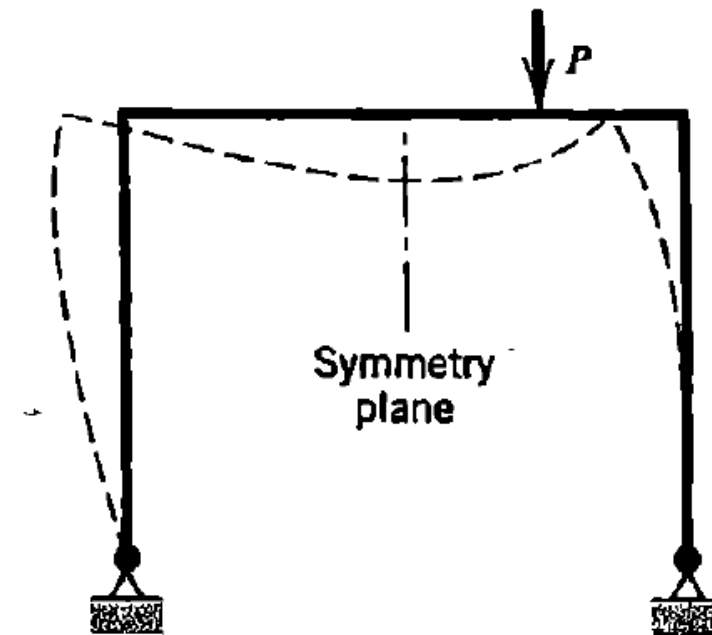
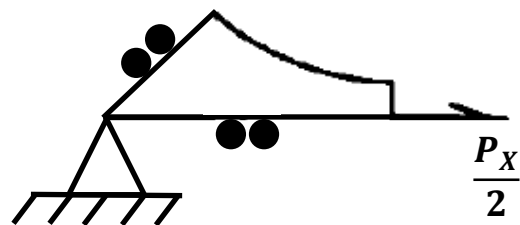
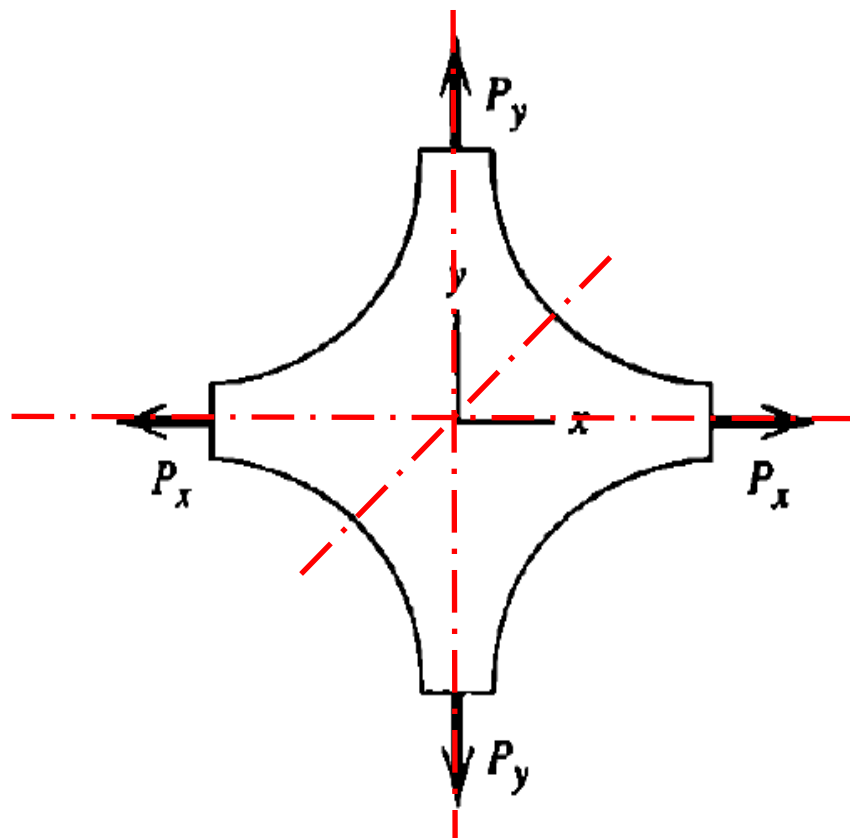


Antisymmetry

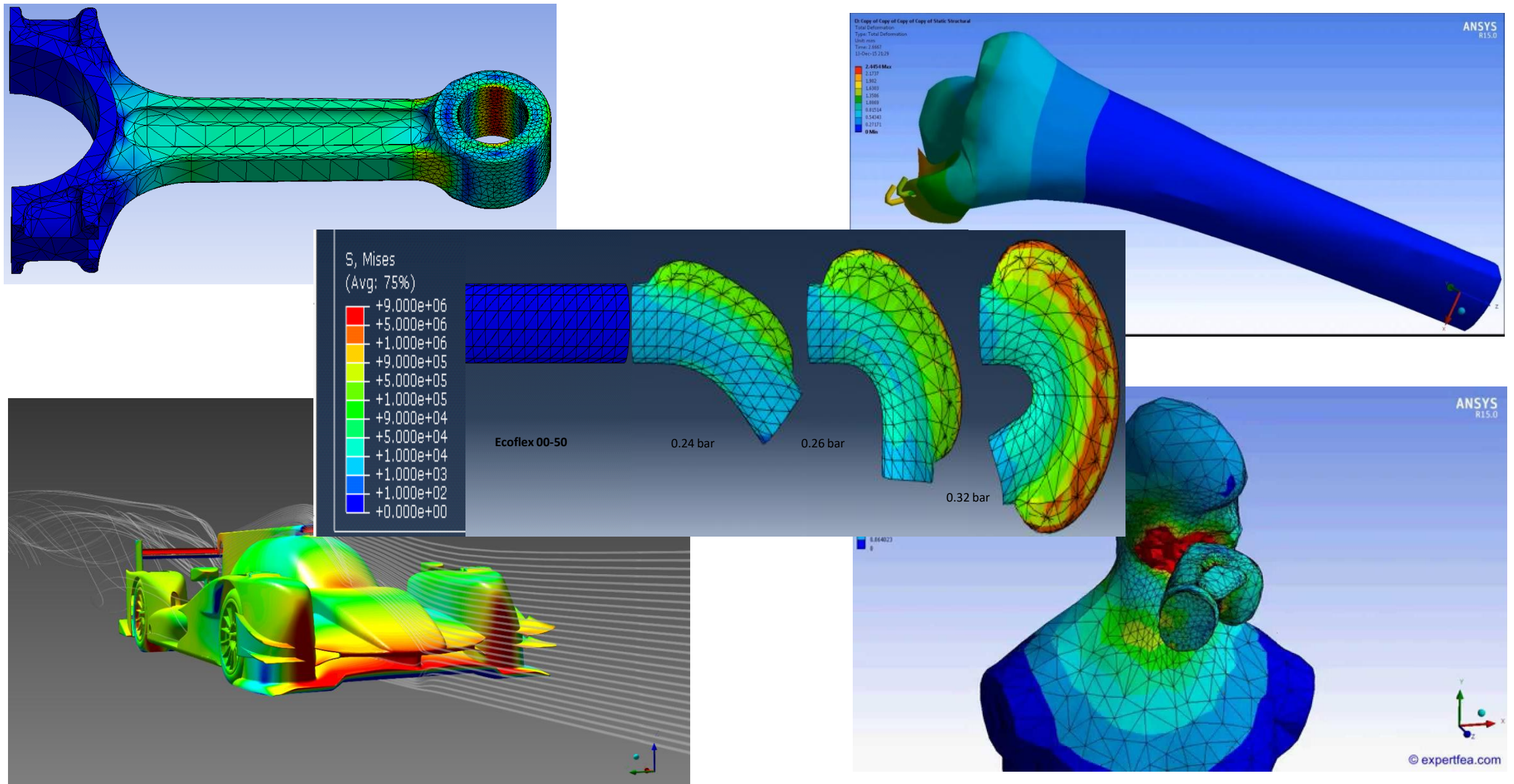
Symmetric geometry and antisymmetric loading



Symmetries



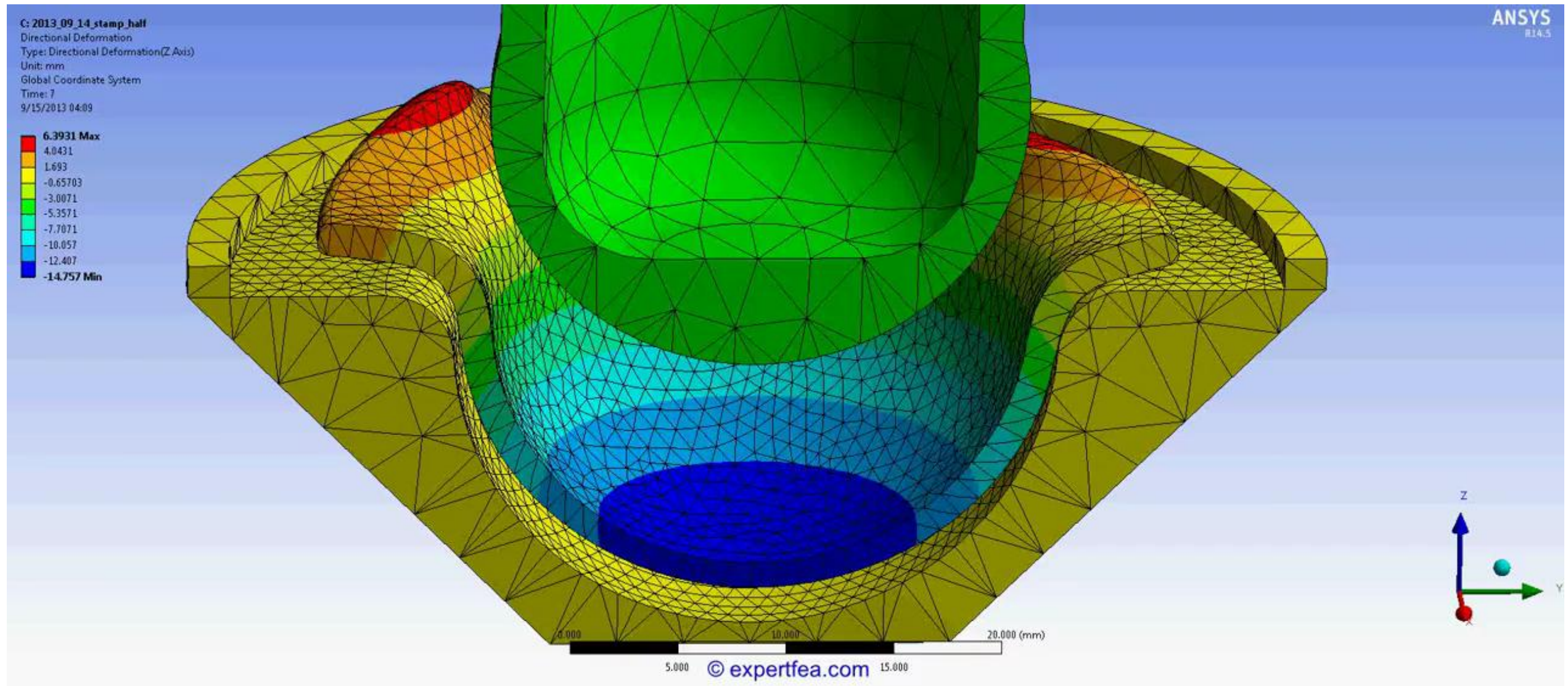
Examples of FEA Analysis



FEA is powerful and widely used in many industries. It is fast (faster than prototyping) and can enable innovation that would be tedious or impossible with any other method, **IF USED CORRECTLY.**



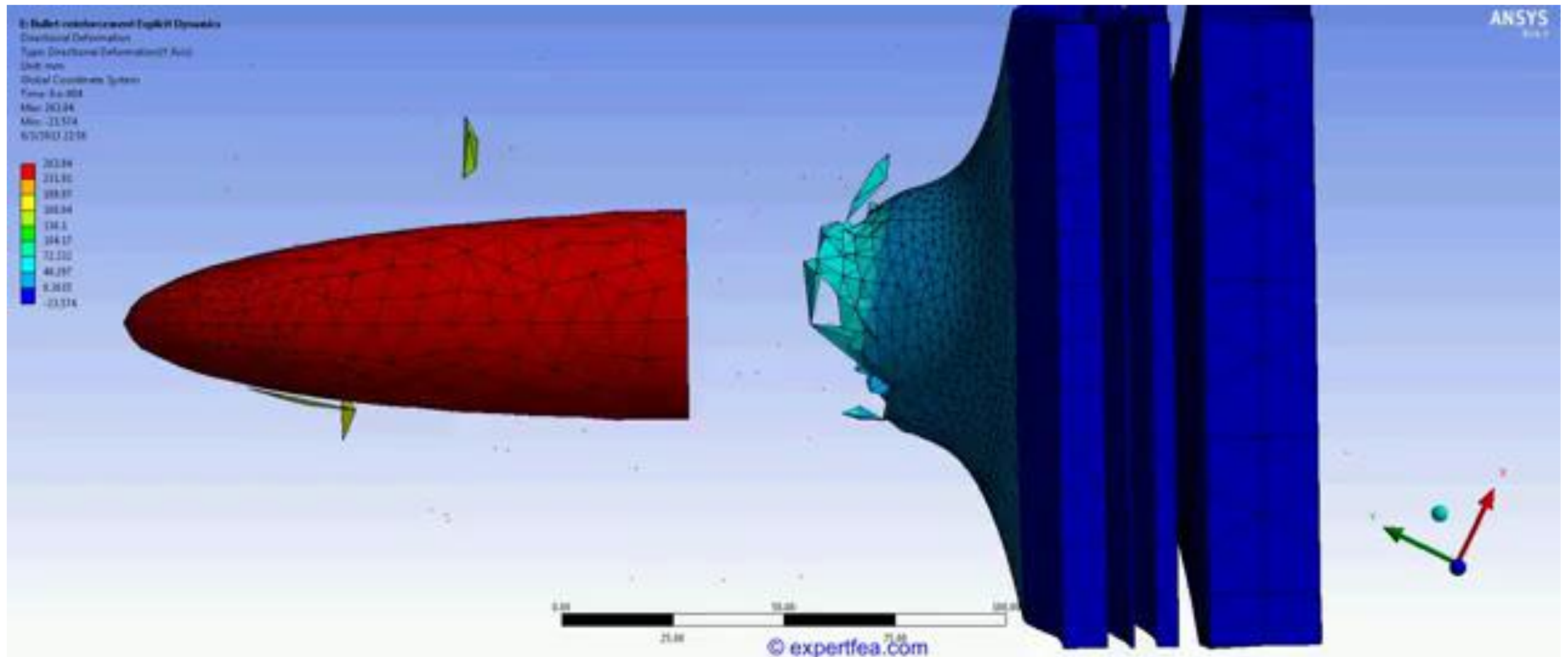
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ANSYS main interfaces and First use of the software

